



ALBUKHARY INTERNATIONAL UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATICS

SEMESTER 3 2024/2025

CCE2333

SOFTWARE TESTING

GROUP PROJECT (20%)

PROJECT TITLE	Usability Testing For the AIU Hup	
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ITEMS	MARKS
Project Report	
Presentation	
Total Mark	

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1.SCOPE

This usability test focuses on the AIU Hub, a web-based electronic platform developed to simplify academic and administrative processes for students at Albukhary International University. Unlike traditional systems, this platform was designed with a student-centered approach, integrating features and services that meet their daily academic needs. The test will cover the following five critical areas:

Registration and login: will users be able to create accounts, be authenticated with a valid account, use invalid credentials, forgotten password, and account recovery.

Course registration: Will users be able to register and drop courses smoothly, be able to take a confirmation slip?

Attendance: We will test how easy it is for users to mark attendance and how easy it is for students to see their attendance records.

Result on screen and reports: We will measure how easy it is for students to check exam results and the ability to show attendance reports.

Profiles: Users will be able to easily change/edit personal information such as name, email address, contact details, and user preferences.

This test focuses on discovering any usability issues that may inhibit user interactions with these components and providing suggestions for actionable improvements to enhance the interface, usability, and responsiveness of the platform to align with students' needs.

2.PURPOSE

The goal of this usability test is to examine whether the AIU Hub provides a satisfactory and effective user experience from the students' perspective. This includes:

Identifying Usability Issues: Identifying the issues a student may face when they are engaging with the various indices in the hub and tasks such as course registration, attendance checking or result checking affect their ability to effectively use the system.

Understanding User Efficiency: Evaluating the usability aspects of the time taken by users to log in to the hub, to register courses, to check attendance, to check results and to manoeuvre between the different sections of the hub.

Understanding Task Completion: Understanding whether students are able to complete the basic academic processes of registering for courses, checking their attendance, updating their profiles, and viewing their results without errors and interruptions.

This test will give detailed information on the strengths and weaknesses of the platform, where improvement will be required, and the AIU Hub platform will address the needs of students and improve their overall academic experience.

3.SCHEDULE AND LOCATION

The usability testing was conducted according to a fixed timetable to cover all the main functions of the AIU Hub. The evaluation took place in Room B09-211, using a structured sequence of activities starting at 10:00 AM and ending at 11:40 AM. A detailed schedule was prepared to assign specific functions to each time slot.

Date	Time	Activity	Venue
15.09.2025	10:00	User registration and login (valid/invalid credentials, password reset)	B09-211
15.09.2025	10:20	Course registration (adding/dropping courses, confirmation slip)	
15.09.2025	10:40	Attendance marking and reviewing attendance records	
15.09.2025	11:00	Profile updates (editing personal and contact details)	
15.09.2025	11:20	Exam results viewing and report generation	
15.09.2025	11:40	Closing session: participant debriefing and feedback collection	

This schedule ensured that all usability aspects of the AIU Hub were tested within the allocated time.

4.SESIONS

Each participant followed the same step-by-step testing procedure lasting 1 hour and 40 minutes. Instead of simply moving from one function to another, the sessions were designed to simulate realistic student tasks, such as logging in, registering for courses, checking attendance, updating a profile, and viewing results. Observers monitored these interactions closely, documenting completion time, errors, and user feedback for every stage.

Session Activity	Duration	Description
Login and Registration	20 minutes	Tested account creation, login with valid and invalid credentials, and password recovery. Observers noted ease of access and error handling.
Course Registration	20 minutes	Students added and dropped courses, then generated confirmation slips. Observers tracked time, success, and usability challenges.
Attendance	20 minutes	Participants marked attendance and reviewed attendance records for clarity and accuracy.
Profile Management	20 minutes	Users updated personal information such as name, email, and contact number to check flexibility of updates.
Results and Reports	20 minutes	Students accessed exam results and generated reports. Observers measured task completion success and efficiency.

At the end of the session, participants gave verbal feedback on their experience, and observations were documented for later analysis.

5.EQUIPMENT AND ENVIRONMENT

The usability testing was carried out in a controlled laboratory environment to ensure consistency and minimize distractions. Standardized equipment and resources were prepared so that all participants experienced the system under the same conditions.

Equipment / Environment	Purpose
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Desktop Computers (Lab PCs)	Each participant used a computer with the AIU Hub preloaded to perform all assigned tasks.
Stable Internet Connection	Ensured uninterrupted access to the web-based platform during testing.
Screen Recording Software	Captured user interactions for later analysis of navigation flow and error patterns.
Observation Materials	Notebooks and digital spreadsheets were used by facilitators to record task completion time, difficulties, and feedback.
Controlled Lab Environment (Room B09-211)	Provided a quiet and distraction-free setting, ensuring participants could focus fully on completing tasks.

6.PARTICIPANTS

The participants for this usability test were carefully chosen to represent the different categories of users who interact with the AIU Hub. The focus group primarily consisted of university students, as they are the main users of the platform. To ensure diversity in perspectives, students from different courses and with varying levels of experience with digital academic systems were included. Each participant engaged with the platform to identify usability issues and provide valuable feedback on user experience.

Type of Participants

Type of Participants	Role	Number of Participants	Description
Students	Testers	13	The test targeted AIU students from various academic programs. They represented different levels of familiarity with online platforms to capture

			a wide range of user experiences and opinions. Participants were directly engaged in testing AIU Hub's key features (login, course registration, attendance, results, and profile updates).
Facilitator	Coordinator	1	Managed the overall usability testing process, including scheduling sessions, briefing participants, and ensuring smooth execution. The facilitator also acted as the main point of coordination between participants and observers.
Note-taker	Coordinator	1	Documented participant feedback, task completion details, and any issues encountered during the sessions. This ensured a structured record of usability findings and insights.

Observers	Coordinator	2	Monitored participants' interactions with the AIU Hub, provided clarification when needed, and recorded real-time behavioral insights. Observers focused on usability challenges and user reactions while maintaining neutrality.
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7.SCENARIO

Scenario 1(Login & Recovery):

The student enters a correct username and password, they are taken directly to the dashboard. If The student enters incorrect information, a pop-up notification appears "Incorrect username or password. The student also tests the "Forgot Password" by Enters their email address or student number and then Receives a message with a password reset link then Enters a new password and successfully logs in.

Objective: To ensure the system is secure and easy to log in, while providing a quick way to retrieve the account.

Scenario 2(Course Registration):

The student opens the registration page and selects a course from the list then the system displays the course information (name, code, hours).Upon confirmation, a "Registration Successful" notification appears, along with an update to the course schedule. The student also can delete a previously registered course. A "Deletion Successful" message appears, and

the record is modified. The student can generate a confirmation slip in PDF format to document the courses registered.

Objective: To ensure that the student can register or drop courses easily and reliably, with an official record.

Scenario 3(Attendance with QR Code):

Objective: To ensure that the QR code attendance system operates accurately, is easy to use for students, and updates the database directly without manual intervention.

Scenario 4(Results & Reports):

The student accesses the results page after the end of the semester. They will see a clear list of the registered courses and final grades. Then will click the "Generate Report" option, and the system will create a PDF file containing the grades and cumulative grade point average (CGPA). The student can also print or download the report for personal use.

Objective: To ensure accurate results, and to generate official reports directly from the system.

Scenario 5(Profile Management):

The student opens the profile page and attempts to edit their information (for example, updating their phone number or email address). After saving, a "Update Successful" notification appears, and the new information appears immediately. They can change their notification settings (email or system notifications). After changing their password and log in again using the new password.

Objective: To ensure that the system allows easy account customization and enhances student security.

8.METRICS

For the usability test of the AIU Hub (StudentHub), we conducted a question-based evaluation aligned with the scenarios defined in our usability testing plan. These questions were used to measure whether participants could successfully complete essential academic tasks. The following subsections detail the metrics, grouped by scenario.

1. For Login & Recovery

Scenario 1: Login & Recovery Testing

❖ Questions:

- Can you log in with a valid username and password?
- Does the AIU Hub display a warning message when invalid credentials are entered?
- Can you recover your account using the "Forgot Password" feature?

❖ Successful Task Completion:

Participants successfully logged in using valid credentials(**Appendix-1**). The system displayed an appropriate error message as invalid credentials when incorrect login information was provided(**Appendix-2**). Password recovery was also effective; students could request a reset link using their student email and log in again with the updated password(**Appendix-3**). A few participants noted minor delays in receiving reset emails, but all were able to regain access.

2. For Course Registration

Scenario 2: Course Registration Testing

❖ Questions:

- Can you register a course successfully and receive confirmation?
- Can you drop a registered course easily?
- Can you generate a printable confirmation slip?

❖ Successful Task Completion:

Most participants successfully registered and dropped courses(**Appendix-5**). Notifications appeared instantly (“Registration Successful” or “Deletion Successful”). Participants could also generate confirmation slips in PDF format. However, some reported that the confirmation slip download took longer than expected during peak usage(**Appendix-7**).

3. For Attendance

Scenario 3: Attendance with QR Code

❖ Questions:

- Can you scan the QR code displayed in class to mark attendance?
- Is attendance updated automatically in the records?
- Can you review your attendance history accurately?
- Does the QR code work after 1 minute ?

❖ Successful Task Completion:

The majority of participants were able to scan the QR code and mark attendance quickly(**Appendix-10**). Records updated immediately in their accounts, showing timestamps of attendance. The QR Code also regenerated after 1 minute(**Appendix-11**). Some participants with older smartphones faced slower QR scanning, but overall, the system worked reliably(**Appendix-12**).

4. For Results & Reports

Scenario 4: Results and Reports Testing

❖ Questions:

- Can you view your semester results clearly?
- Can you generate a cumulative report in PDF format?
- Is the report accurate and downloadable?

❖ Successful Task Completion:

Participants successfully accessed their grades and generated official reports containing CGPA values(**Appendix-13**). Most found the process smooth, though a few participants mentioned that large report files took longer to load. The accuracy of data (course codes, grades, GPA) was confirmed by all testers(**Appendix-14**).

5. For Profile Management

Scenario 5: Profile Management Testing

❖ Questions:

- Can you edit and save profile details such as email and AIU ID ?
- Can you update security credentials (password)?
- Can you adjust notification preferences successfully?

❖ Successful Task Completion:

Profile updates (personal info, email, password) were successful for most participants. Password changes were effective, with immediate access using the new credentials(**Appendix-15**). Notification preferences (system/email) could also be toggled successfully(**Appendix-16**). However, a few participants noted that the confirmation messages for updates were delayed by several seconds

Critical Errors

Critical errors are major issues that prevent participants from completing tasks or achieving scenario goals as intended. These errors directly affected usability and blocked essential functions of the AIU Hub.

- **For Scenario 2: Course Registration Testing**

- **Delayed Confirmation Slip Download**

Some participants were unable to generate or download confirmation slips during peak usage times. This failure prevented them from securing official proof of course registration.

- **For Scenario 3: Attendance with QR Code**

- **QR Code Scanning Issues on Older Devices**

A few participants using older smartphones could not scan the QR code successfully within the time limit, which resulted in incomplete attendance records.

- **For Scenario 4: Results & Reports Testing**

- **Slow Loading of Large Reports**

Participants attempting to generate full semester reports faced long

loading times, leading to incomplete downloads and an inability to confirm CGPA values in real-time.

Non-Critical Errors

Non-critical errors did not stop participants from completing tasks but negatively impacted efficiency, speed, or user satisfaction.

- **For Scenario 1: Login & Recovery Testing**
 - **Delayed Password Reset Emails**

A few participants noted delays in receiving password reset links. While they were eventually able to log in, this slowed the recovery process.
- **For Scenario 2: Course Registration Testing**
 - **System Lag During Peak Hours**

Course addition and deletion sometimes took longer than expected to process, which caused confusion about whether the action was successful.
- **For Scenario 5: Profile Management Testing**
 - **Delayed Confirmation Messages**

Updates to personal details and notification preferences were applied successfully, but confirmation alerts appeared several seconds late, leaving participants uncertain about whether changes were saved.

Error-Free Rate

- **Participants Completing Tasks Without Any Errors:** 5 out of 13
- **Error-Free Rate:** 38.4615%

This indicates that only 38.4615% of participants completed all scenarios without encountering any critical or non-critical errors. While the AIU Hub platform is functional and supports essential student activities, there are still areas requiring improvement, particularly in system responsiveness, error handling, and device compatibility.

Task on time:

No.	Date	Time Stamp	Name	Time Taken
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1	16/09/2025	10:00:00	M. Arif Ardelian	47 minutes
2	16/09/2025	10:03:00	Mahmud	1 hour 2 minutes
3	16/09/2025	10:06:00	Habiba	55 minutes
4	16/09/2025	10:09:00	Akash	50 minutes
5	16/09/2025	10:12:00	Moon	1 hour 5 minutes
6	16/09/2025	10:15:00	Somayaa	53 minutes
7	16/09/2025	10:18:00	Sagor	1 hour
8	16/09/2025	10:21:00	Khadiga	58 minutes
9	16/09/2025	10:24:00	Ahad	1 hour 8 minutes
10	16/09/2025	10:27:00	Angel	54 minutes
11	16/09/2025	10:30:00	Basma	1 hour 7 minutes
12	16/09/2025	10:33:00	Baraa	52 minutes
13	16/09/2025	10:36:00	Doha	1 hour 3 minutes

9.0 QUANTITATIVE MATRICS

Tester	Successful task completion	Critical errors	Noncritical errors	Errorfree rate	Time on task	Likes, dislikes
M. ARIF ARDELIAN	4 out of 20	3	1	80%	50 minutes	Liked: easy to navigate Disliked: A bit slow and cluttered—needs

						cleaner, faster design.
Ahad	3 Out of 20	2	1	85%	45 Minutes	Liked: Noti fication Disliked: Disappeared content
Habiba	7 Out of 20	5	2	65%	50 Minutes	Liked: its good Disliked: Server Down
Khadiga	4 out of 20	3	1	80%	1 Hour	Liked: Have many functions Disliked: Sometimes it delays and not working well
Baraa	8 out of 20	6	2	60%	1 Hour	Liked: Help us manage differences things in one place Disliked: Need to be easy to use
Basma	6 out of 20	5	1	70%	1 Hour	Liked: Nothing Disliked: accessibility

Akash	1 out of 20	1	0	95%	1 Hour	Liked:Ove rall Good Disliked: Overall Good
Sagor	4 out of 20	3	1	80%	50 Minutes	Liked: Attendance system Disliked: Server Down sometimes
Doha	4 out of 20	3	1	80%	55 Minutes	Liked: Easy to use Disliked: Nothing
Somayaa	4 out of 20	2	2	80%	1 hour	Liked: Easy to use Disliked: Nothing
Mahmud	4 out of 20	4	0	80%	50 Minutes	Liked: Course Registration is very easy Disliked: Server down often
Angel	6 out of 20	6	1	70%	1 hour	Liked: everything is organized Disliked: Server down

						during exam week
Moon	5 out of 20	4	1	75%	55 minutes	Liked: Attendance Disliked: Attendance sometimes show more than 100%

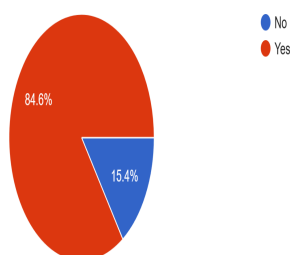
10. ROLES

Name	Role
Khadiga	Scope, Purpose, Report Formatting
Wejdan	SCHEDULE AND LOCATION, SESSIONS, EQUIPMENTS
Mehedi	Participants, Quantitative Metrics
Baraa	Scenarios
Rafi	Metrics

11. APPENDICES

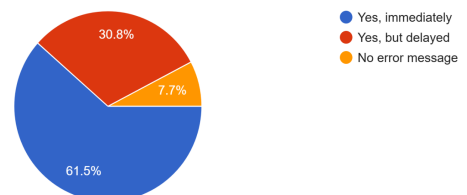
Were you able to log in successfully using a valid username and password?

13 responses



When entering incorrect credentials, did the system display an error message?

13 responses



Appendix: 2

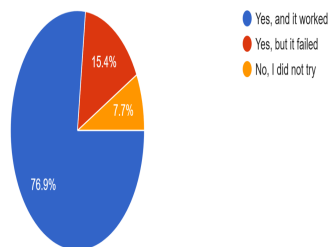
Overall, how easy was the login & recovery process?

13 responses



Appendix: 1

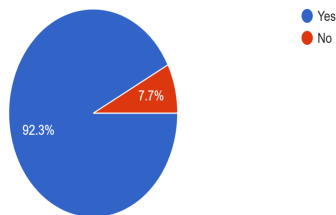
Did you attempt to recover your account using the "Forgot Password" feature?
13 responses



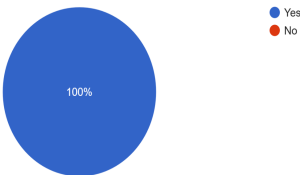
Appendix: 4

Appendix: 3

Were you able to register a course successfully?
13 responses



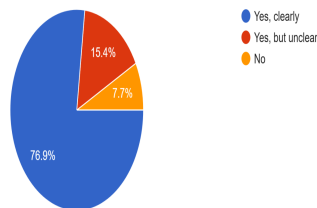
Were you able to drop a course easily?
13 responses



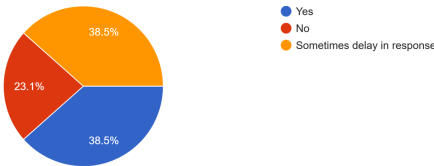
Appendix: 6

Appendix: 5

Did the system display confirmation messages (e.g., "Registration Successful")?
13 responses



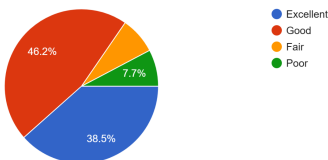
Do the server got down in peak time
13 responses



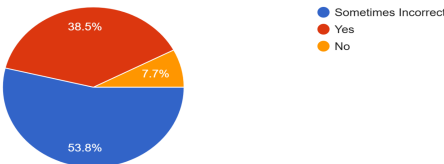
Appendix: 8

Appendix: 7

How would you rate the overall course registration experience?
13 responses



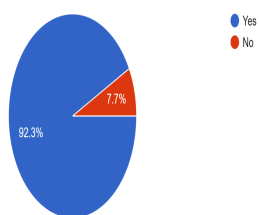
Were you able to view your attendance history accurately?
13 responses



Appendix:9

Did the QR code regenerate after 1 minute as expected?

13 responses

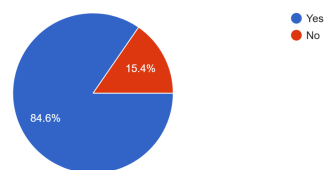


Appendix:11

Appendix:10

Were you able to scan the QR code to mark attendance?

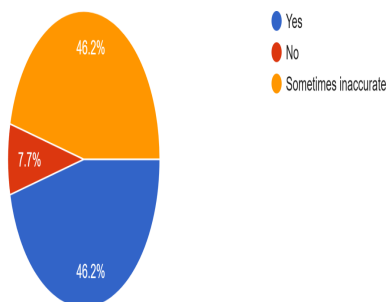
13 responses



Appendix: 12

Was the report accurate (course codes, grades, GPA)?

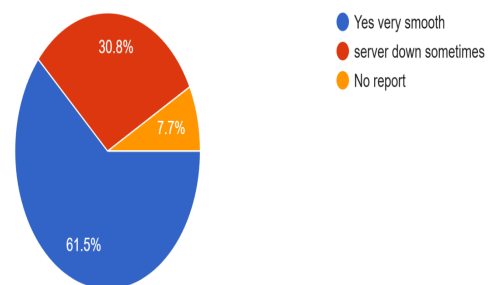
13 responses



Appendix:13

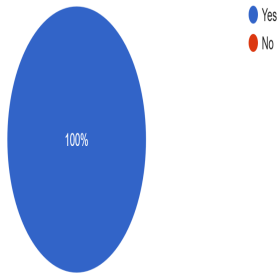
Were you able to download the report successfully?

13 responses



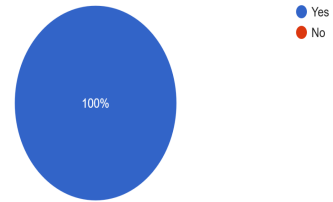
Appendix: 14

able to update your password successfully?
15 responses



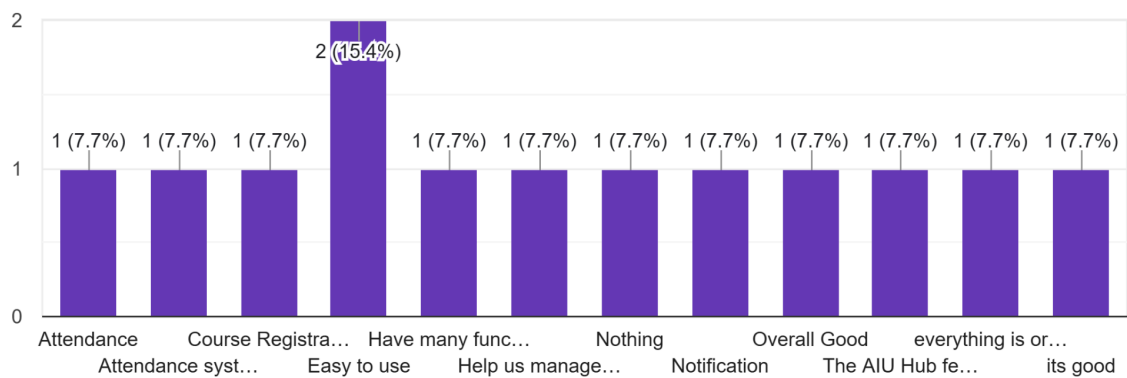
Appendix: 15

Did your new password work immediately after updating?
13 responses



Appendix: 16

What did you like most about the AIU Hub?
13 responses



Appendix: 17



Figure: 1